

Integrated Climate-intelligent Salt Production Ecosystem: A Multi-modal AI Framework For Sustainable Yield Optimization In Tropical Coastal Environments

Project ID: 25-26J-431

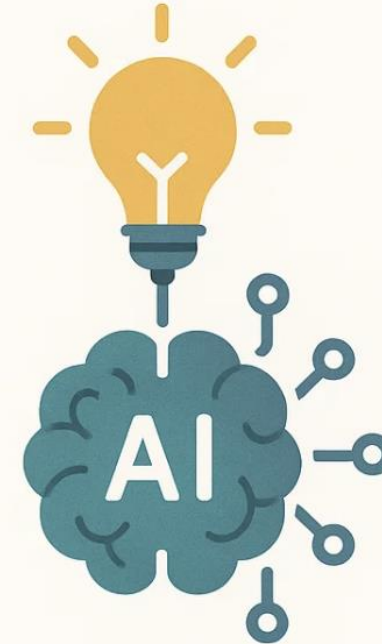
Supervised by: Dr. Nathali Silva
Co-supervised by: Mrs. Thilini Jayalath

Integrated Climate-Intelligent Salt Production Ecosystem



Research Problem

Climate variability causes unpredictable salt yields, economic losses, and quality issues.



Research Question

How can an AI-driven, multi-modal forecasting and planning system improve resilience, optimize yields, and support sustainable salt production in Sri Lanka?

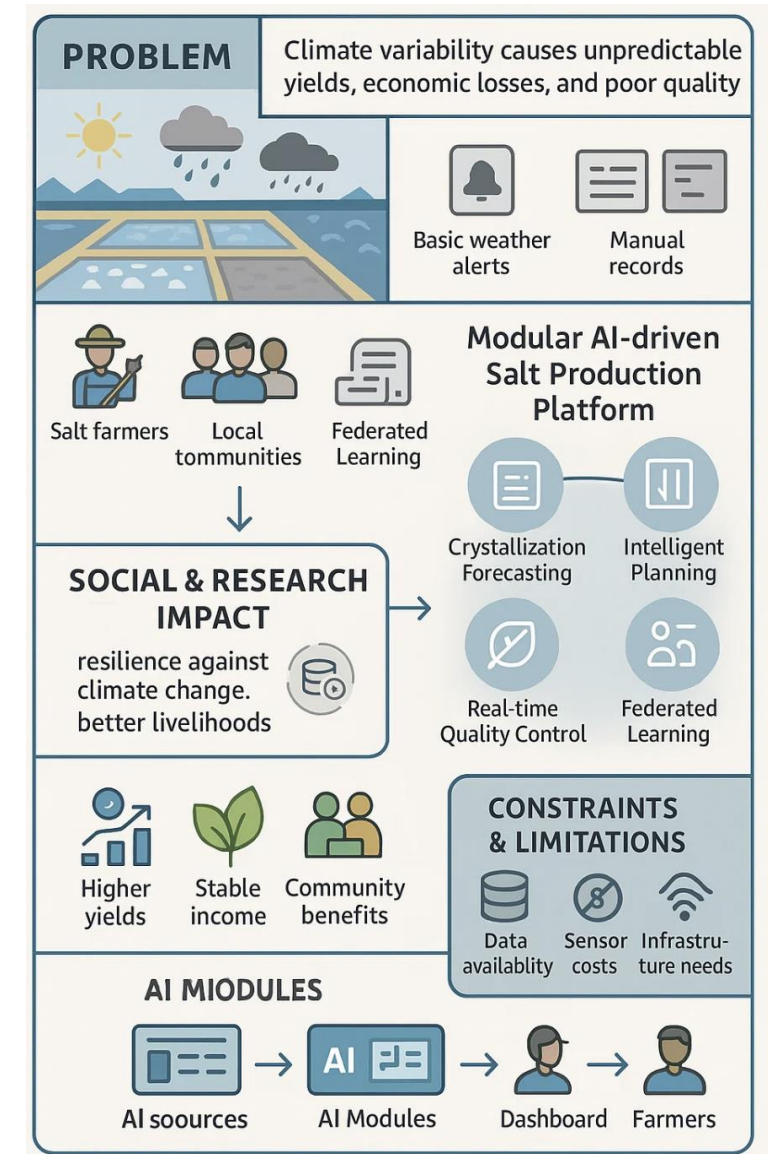
Integrated Climate-Intelligent Salt Production Ecosystem

Research Objectives:

Build a flexible AI system that:

- Predicts environmental conditions for salt making (crystallization forecasts)
- Supports smart planning using multiple important factors
- Automatically monitors salt quality in real time with computer vision
- Helps manage and reuse salt waste efficiently across different locations using federated learning

The **goal** is to improve productivity, sustainability, and climate resilience for salt producers of all sizes.



Integrated Climate-Intelligent Salt Production Ecosystem

Social & Research Impact

- Strengthens resilience & sustainability of Sri Lanka's salt industry
- Stabilizes production, jobs & exports
- Introduces AI-driven planning for tropical coastal salt sectors

Research Gaps

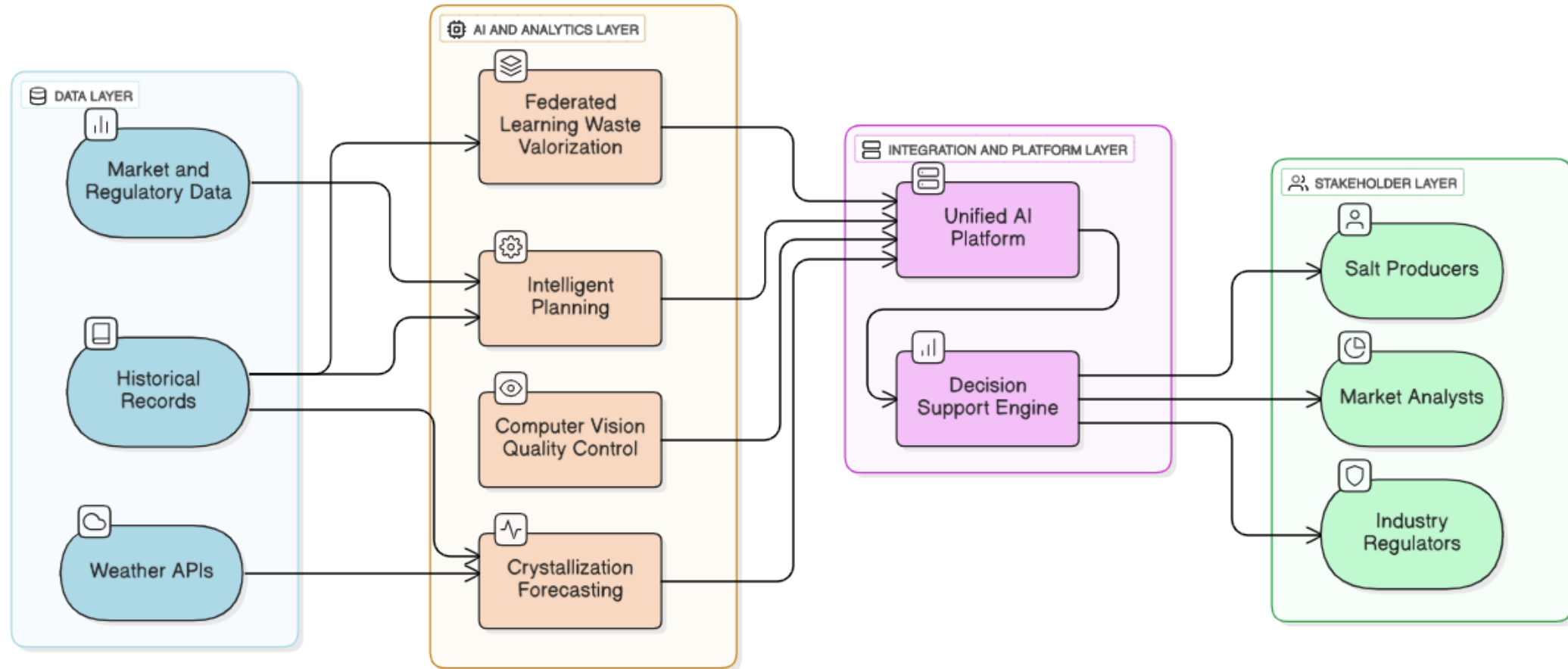
- Current planning is manual & fragmented
- AI applied in agriculture/fisheries, not salt industry
- Tools lack dynamic adaptation & stakeholder focus

Constraints

- Relies on accurate data & digitization
- Needs stakeholder training & adoption
- Internet reliability is essential
- Continuous updates & feedback required

Integrated Climate-Intelligent Salt Production Ecosystem

- System diagram



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IT22308016 | SIRIMANNA R.D.I.B

Crystallization Forecasting

Software Engineering

Crystallization Forecasting

Gap: No accurate system to predict when salt is ready for harvest.

Problem

- Relies on manual averages or basic weather alerts
- Cannot predict the best harvest time for different salt crystals

Shortcomings

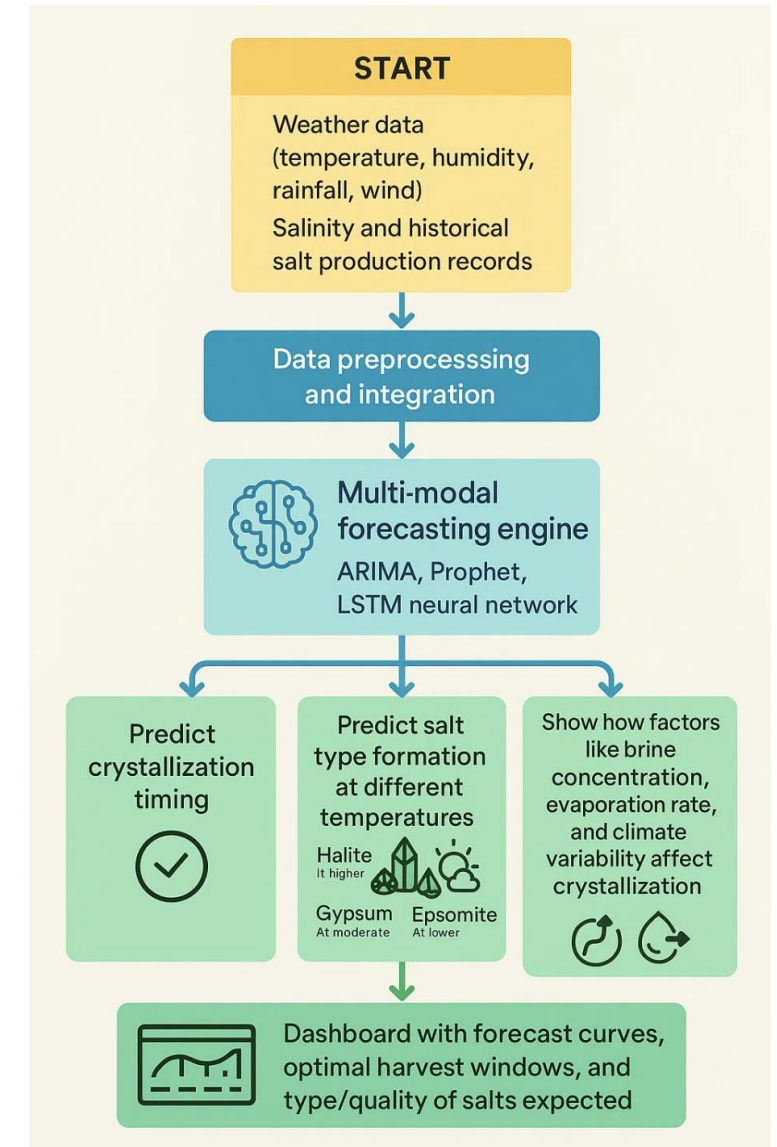
- Only simple weather updates, no full data integration
- No smart forecasting for harvest planning

Solution : AI-powered Forecasting System using:

- Weather, temperature, salinity, and records
- Advanced models (ARIMA, Prophet, LSTM)

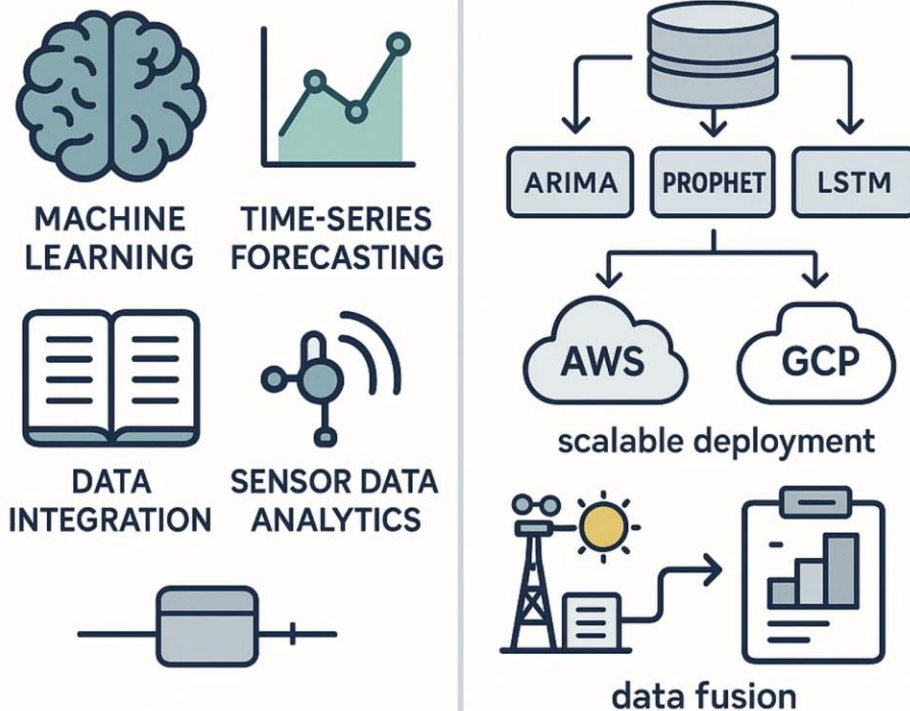
Benefits

- Combines all data for precise forecasts
- Gives early, reliable harvest timing
- Reduces risks and improves planning



Crystallization Forecasting

DOMAINS OF SPECIALIZATION & TECHNOLOGIES



- **Domains of Specialization-based Knowledge:**
 - Machine learning
 - Time series forecasting
 - Data integration
 - Sensor data analytics.
- **Technologies:**
 - ARIMA, Prophet, and LSTM models for time series forecasting.
 - Data fusion from weather stations and manual records.
 - AWS / TensorFlow cloud platforms for scalable model training and deployment.

Crystallization Forecasting

Evaluation

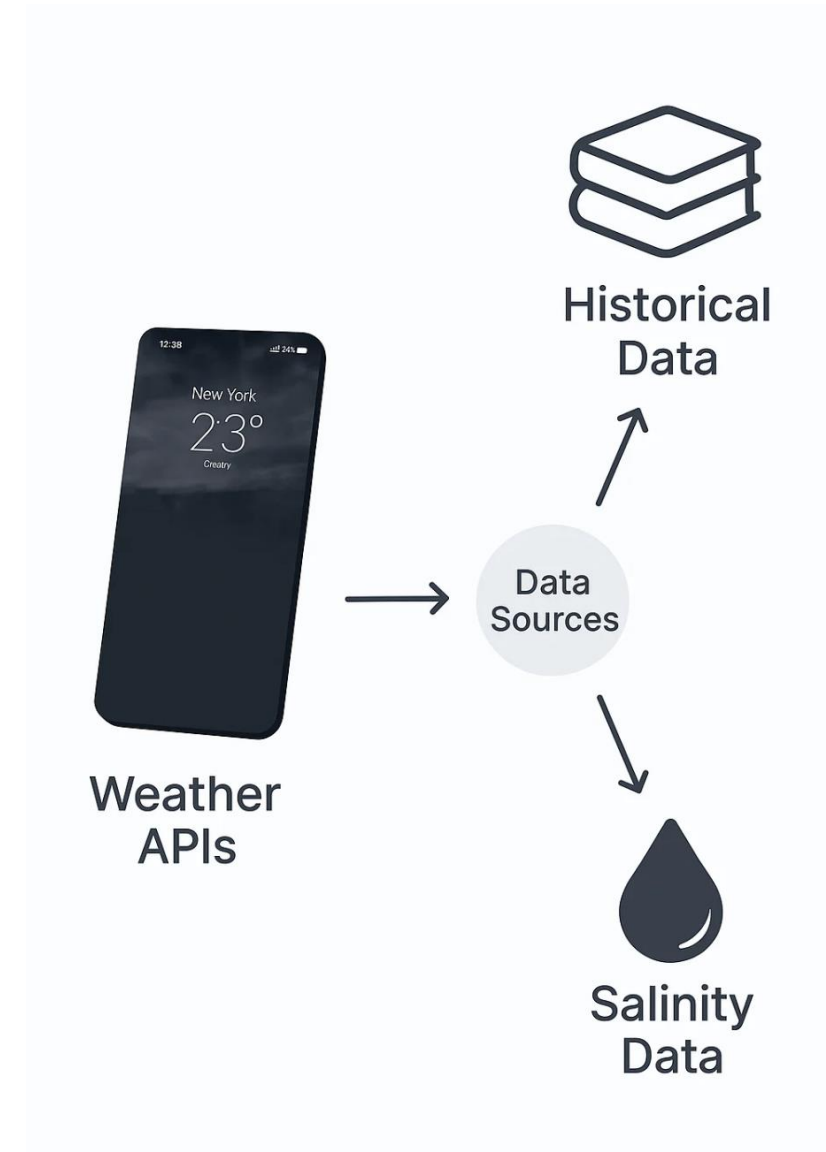
- Test on past weather & production data
- Pilot trials with saltern operators

Data Availability

- Historical data digitized & standardized
- Real-time updates via weather APIs

Ethical Clearance

- Data collected with consent (Puttalam Salt Society)
- Follows privacy & security standards



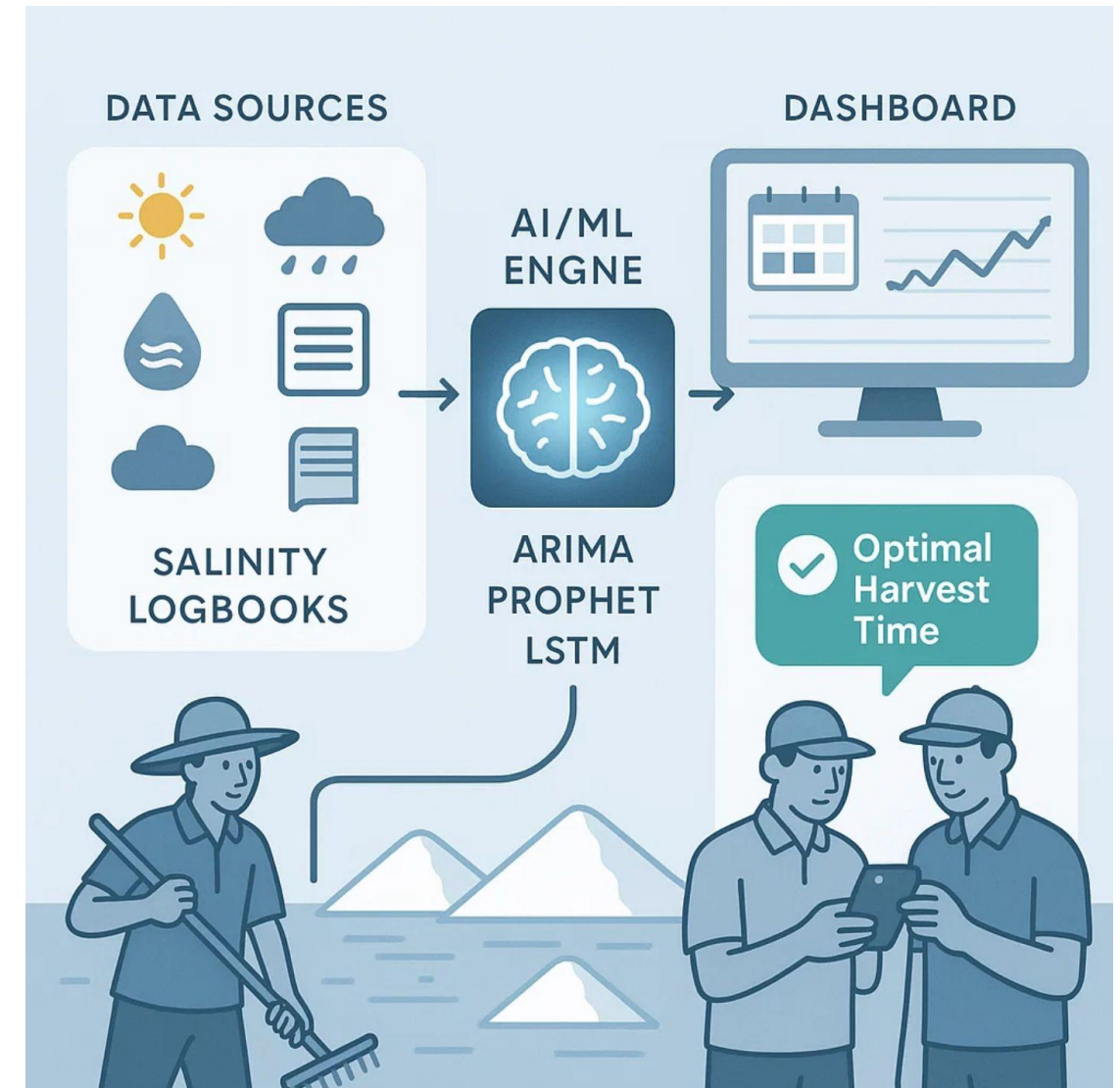
Crystallization Forecasting

System Overview

- Automated pipeline collects weather, sensor, and production data
- AI models forecast crystallization and harvest timing
- Web dashboard shows predictions & sends harvest alerts

Real-World Usage

- Early notice of best crystallization & harvest periods
- Helps producers plan resources, reduce losses, and boost yield





IT22893734 | ANSAR T. D

Intelligent Planning Engine

Software Engineering

Intelligent Planning Engine

Gap: No real-time, AI-driven planning system exists for the Sri Lankan salt industry.

Problem: Current planning is:

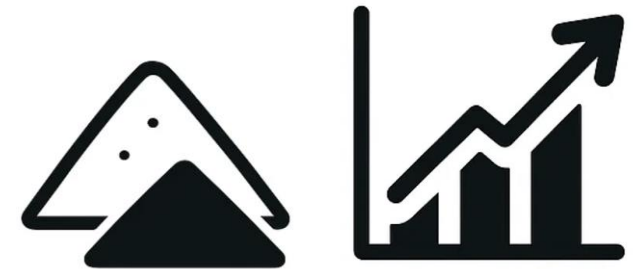
- Reactive and manual | Fragmented | Poor at integrating environmental, economic, and regulatory factors

Shortcomings

- Decisions rely on incomplete or delayed data
- No ability to adapt to real-time environmental or market changes

Proposed Solution: Adaptive Intelligent Planning Engine using,

- Multi-parameter optimization
- Reinforcement learning
- Fuzzy logic

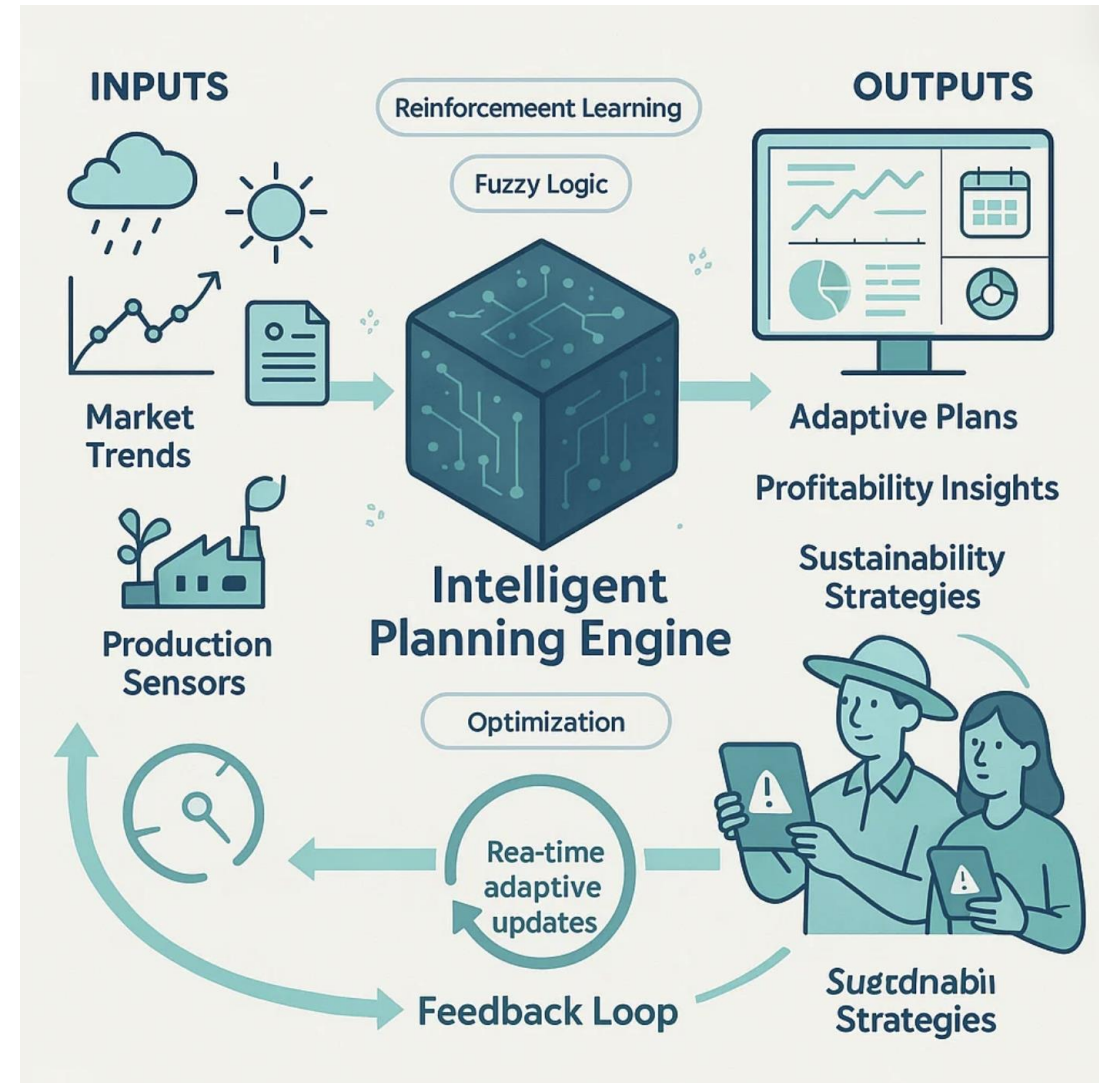


How It Helps

- Integrates real-time environmental forecasts, market data, production logs, and regulations
- Continuously updates and customizes plans based on user priorities

Intelligent Planning Engine

- **Domains of Specialization-based Knowledge:**
 - AI-driven decision support
 - multi-parameter optimization
 - reinforcement learning
 - fuzzy logic
 - cloud-based modular architecture
- **Technologies:**
 - Rule-based Optimization - experta (Python library for building expert systems)
 - Reinforcement Learning - TensorFlow (TF-Agents)
 - Fuzzy Logic Models - scikit-fuzzy (Python fuzzy logic toolbox)
 - AWS Cloud computing infrastructure and scalable data pipelines
 - MongoDB



Intelligent Planning Engine

Evaluation

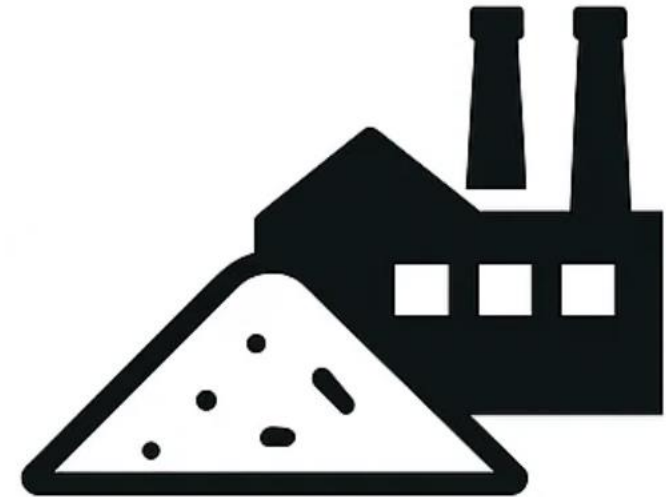
- Simulations with past market, weather & production data
- Pilot trials with salt producers (profitability & compliance)
- User feedback to improve the system

Data Availability

- Weather, market, production & regulation data
- Verified and digitized from multiple sources
- Production records from Puttalam Salt Society

Ethical Clearance

- Stakeholder approvals for data use from the Puttalam salt society
- Secure, role-based access to sensitive data



Intelligent Planning Engine

System Overview

- Web-based AI planning engine
- Interactive dashboards & scenario analysis
- Uses multiple data sources for optimized plans

Real-World Usage

- Managers set priorities (profit, quality, sustainability)
- System creates & updates plans dynamically
- Helps handle climate changes & market shifts





IT22346322 | ARSHAQ M.J.M

Computer Vision Quality Control System

Software Engineering

Computer Vision Quality Control System

- **Gap:** No integrated AI system for real-time salt quality inspection in Sri Lanka.
- **Creative Solution:** AI-powered Computer Vision System using CNNs for automated impurity detection and multi-parameter grading, for proactive quality control.
- **Problem Definition:** Manual inspections in salterns are subjective and time-consuming, failing to detect minute impurities, leading to higher rejection rates.
- **Shortcomings:**
 - Subjective and error-prone manual checks miss subtle defects.
 - No multi-parameter evaluation (limited to basic visual inspection).
- **How Proposed Solution Addresses:**
 - Automates detection using CNN
 - Provides instant feedback via high-res imaging and dashboard alerts.
 - Evaluates whiteness, crystal size, and uniformity simultaneously.

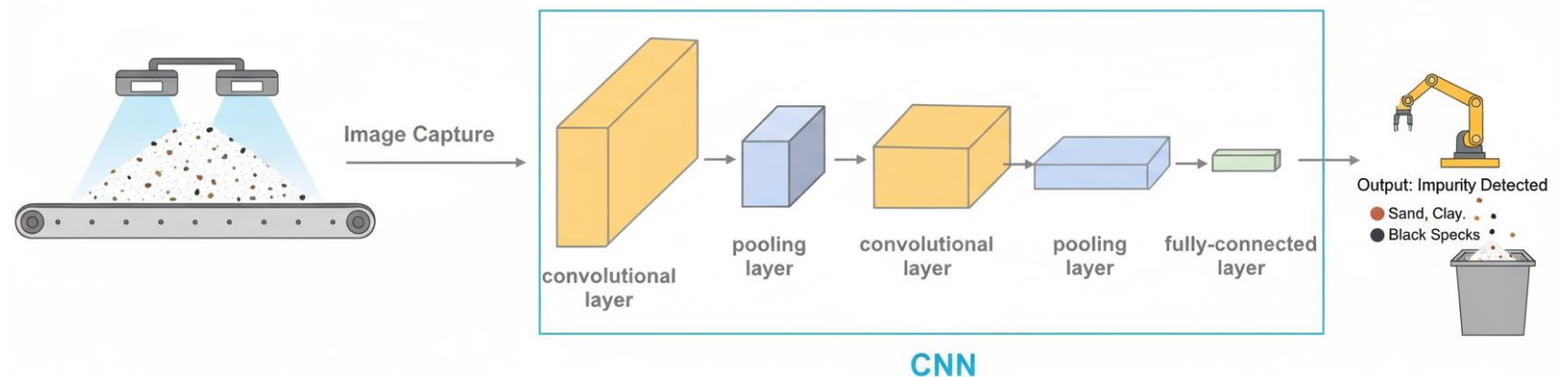
Computer Vision Quality Control System

- **Domains of Specialization-based Knowledge:**

Modular design integrating image acquisition, CNN-based processing, with scalable deployment

- **Technologies:**

- MobileNetV2 Convolutional Neural Networks (CNNs)
- TensorFlow.js Real-Time Image Processing
- LabelImg/CVAT for dataset creation



Computer Vision Quality Control System

- Evaluation:
 - Use cross-validation on annotated datasets
 - Pilot testing in Salterns
 - compliance with SLSI/ISO/Codex
- Data Availability:
 - Salt sample images from Puttalam salterns
 - Augmented datasets for training
- Ethical Clearance:
 - Informed consent from saltern owners
 - Data anonymized with Privacy preserved

Computer Vision Quality Control System

- System Overview:
 - AI-powered Computer Vision System for real-time salt quality inspection
 - Uses CNN-based impurity detection & multi-parameter grading
 - Integrated web-based dashboard for operators
- Real-World Usage:
 - Install cameras & LED lights on salt conveyor belts
 - System captures live images → analyzes via CNN model → grades salt quality
 - Ensures compliance with SLSI, Codex, and ISO 22000 standards





IT22354310 | PERUMBULI P.G.R.M.D

Federated Learning for Waste Valorization

Software Engineering

Federated Learning for Waste Valorization

Gap:

- No privacy-preserving, collaborative AI solutions for industrial salt waste recovery.

Creative Solution:

- Federated AI with differential privacy for secure, decentralized optimization.

Problem Definition:

- Lack of collaboration due to proprietary data concerns.
- Centralized AI leads to inefficiency and risk.

Shortcomings of Existing Systems:

- Do not optimize waste reuse across multiple facilities.
- Fail to protect sensitive process data.

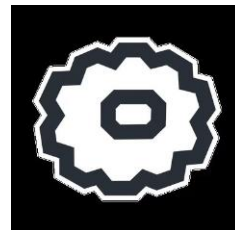
Federated Learning for Waste Valorization

Domains of Knowledge:

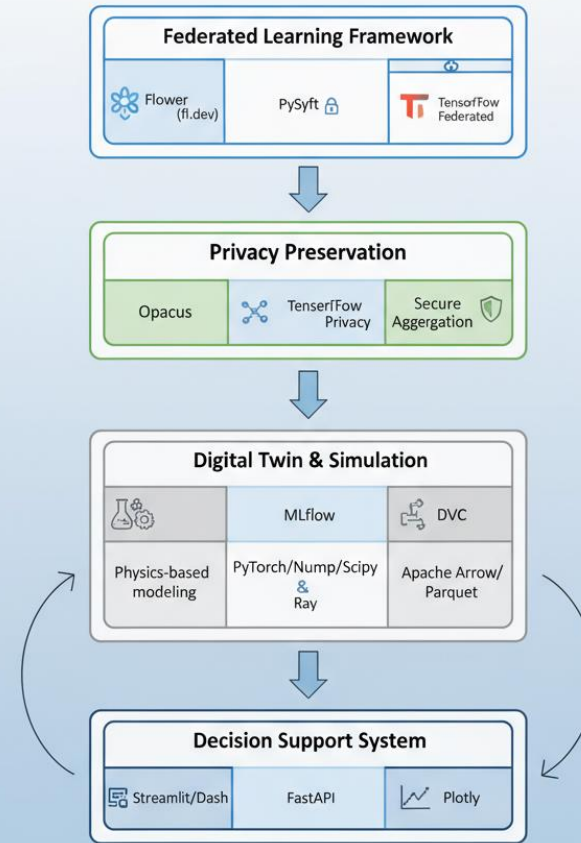
- Federated Learning (distributed AI training)
- Digital Twin simulations
- Decision Support Systems

Technologies:

- TensorFlow Federated / Flower/ Pysyft
- Opacus (privacy) / TensorFlow(privacy)
- Simulation: Unity / MATLAB



Technology Stack for Federated Waste Valorization



Federated Learning for Waste Valorization

Evaluation:

- Accuracy of predictions (crystallization, yield optimization)
- Waste reduction %
- Resource recovery improvements

Data & Ethics:

- Physics-informed synthetic data + pilot plant data
- Privacy preserved (differential privacy + secure aggregation)



Federated Learning for Waste Valorization

System Overview:

- Multiple saltern sites run local AI models.
- Federated learning aggregates insights without sharing raw data.
- Decision Support System provides actionable optimization.

Real-World Use:

- Salt producers get waste recovery recommendations.
- Enhances resource efficiency & compliance with environmental laws.
- Improves profitability through by-product valorization.

Integrated Climate-Intelligent Salt Production Ecosystem

Commercialization

- AI-driven, sustainable salt production platform
- Scalable SaaS model with strong demand

Investment & Pricing

- Cost: LKR 172,000
- Tiered subscriptions + setup/consulting fees

Target Market

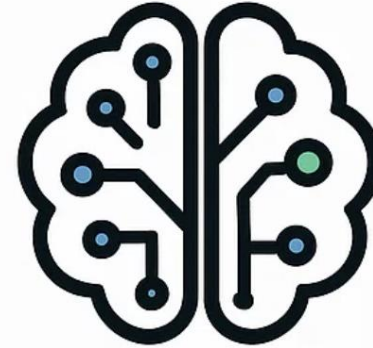
- Salt producers (small to large)
- Govt. & export stakeholders in tropical regions

Value Proposition

- Higher yield, quality & profit
- Real-time planning + compliance support

Intellectual Property

- Proprietary AI algorithms & platform design





Thank you.

It's time for Q&A